

Week 6 Weekly Report

Week 6 trained and compared the two ML approaches. The direct-price revision replaces unbounded linear regression with bounded normalized time-value regression. Purged training CV tunes its parameters; the 252-row historical test is reused for retrospective comparison after the live failure.

1 Work Completed

- Retained the original volatility-route candidate comparison. Repaired direct pricing through regularization, normalized time value and constrained moneyness tails, without adding dividend-growth features.
- Retained validation-based selection for Approach 1; tuned the repaired direct model with purged training cross-validation.
- Evaluated BSM, ML volatility plus BSM, and direct supervised pricing using MAE, RMSE, R^2 , and mean error.
- Measured results by VIX regime to identify unstable periods.
- Generated SHAP and permutation-importance outputs for the exact selected models.
- Saved the selected model bundles and metadata for Week 7 and Week 8 reuse.

2 Deliverables

- Week6_Model_Training_Comparison.ipynb
- model_training.py
- training_config.json
- trained_models/best_volatility_model.pkl and trained_models/best_direct_pricing_model.pkl

joblib runtime copies and model_bundle_metadata.json

- MODEL_CARD.md
- metrics, predictions, SHAP tables, and figures
- Model Performance Report

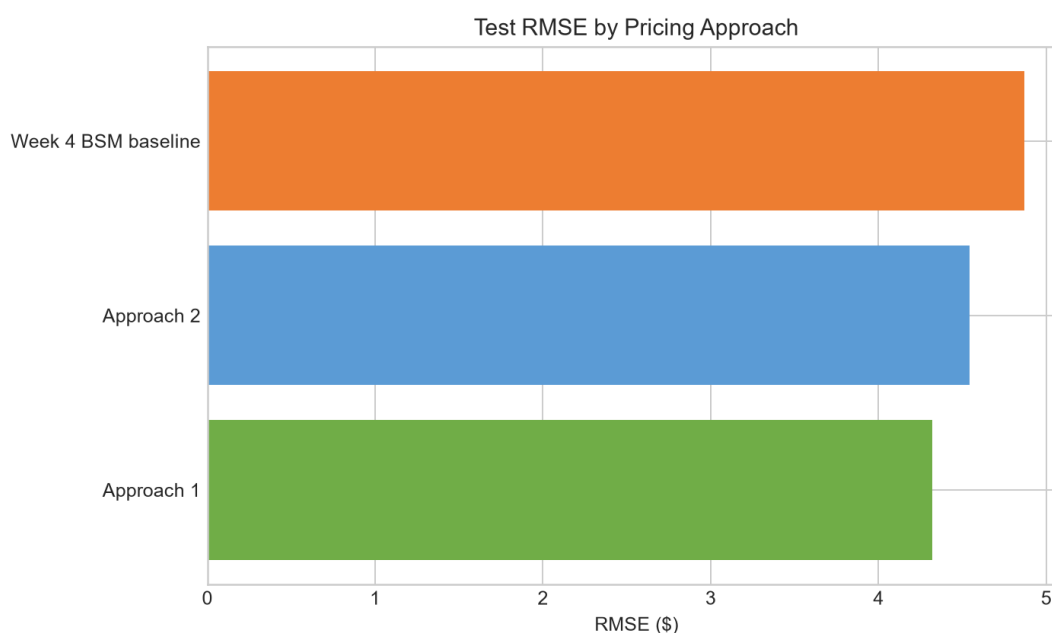
3 Progress Summary

Approach 1 retains the lowest historical test RMSE, 4.3203, versus BSM 4.8706 and repaired Approach 2 4.5404. Direct test MAE improves to 2.8501. Its validation RMSE worsens from 5.4837 to 7.3983, so the revision is not uniformly more accurate. All test comparisons use the same proxy targets and dates.

4 Experimental Results

Pricing route	MAE	RMSE	R ²	Role
Approach 1 RF vol plus BSM	3.1384	4.3203	0.9575	Selected
Week 4 BSM baseline	3.5698	4.8706	0.9460	Benchmark
Approach 2 direct price	2.8501	4.5404	0.9530	Bounded revision

Figure 1 Historical test RMSE comparison



5 Issues Encountered

- The test target is an ex-post forward-volatility chooser proxy, so improved proxy error does not prove improved performance against actual chooser transactions.
- The legacy direct model produced a negative live estimate. The replacement predicts within structural price bounds without zero clipping. Extreme-moneyness tests and pickle round trips passed. Out-of-range warnings remain because valid bounds do not establish market accuracy.
- SHAP values describe model behavior on the analyzed sample; they do not establish causal effects.

6 Plan for Week 7

- Run the required volatility and rate stress tests.
- Quantify the influence of VIX-related features.
- Build the public-data update path and Streamlit prototype.